

A Gaussian Rank-One Obstruction to an Integrable-Spreading Entropy Formula

Solution packet for arXiv:2603.23744

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Status. `candidate_counterexample_likely_valid`. This is a full counterexample to the natural formulation based only on an integrable spreading function, even with arbitrarily small spreading norm. It is not a counterexample to a stricter compact-spreading-support formulation or to a scaled Szegő-limit theorem.

1 Source question and precise scope

Allard and Bölcskei study entropy of compact hypoelliptic pseudodifferential operators and prove phase-space log-integral asymptotics under negative-order hypoellipticity and regular variation [1]. On printed page 6 they write:

“Developing a rigorous analogue of Theorem 1 for underspread operators is an interesting direction for future work.”

The preceding sentence anticipates entropy formulae of the same log-integral form. The official TeX source’s commented development (lines 1078–1098 of the included `source_2603.23744.tex`) makes one contemplated formulation precise: the spreading function, the symplectic Fourier transform of the Weyl symbol, has finite L^1 norm.

work is to identify precise conditions under which the insertion of (8) into the entropy and minimax-risk integrals can be rigorously justified in the compact, symbol-decay setting considered here.

The volume principle underlying (8) is reminiscent of the approximate diagonalization results in underspread operator theory [8, 22, 30, 37, 44, 45, 48], in that both frameworks interpret the Weyl symbol as an effective time–frequency transfer function and relate spectral quantities to its pointwise values through a phase-space localization principle. In that setting, one expects entropy formulae of the same log-integral form as (9), paralleling Shannon-type expressions for frequency-selective channels. Developing a rigorous analogue of Theorem 1 for underspread operators is an interesting direction for future work.

We now present a reformulation of Theorem 1 that makes the connection between entropy and the symbol explicit. Again, the symbol σ in Corollary 2 may be taken to be the left, right, or Weyl symbol of the associated operator.

We refute exactly the assertion that this integrable-spreading hypothesis—or even arbitrary smallness of that norm—is sufficient for

$$H_T(\varepsilon) \sim \int_{\mathbb{R}^{2d}} \log_+ \left(\frac{a(z)}{\varepsilon} \right) dz, \quad \varepsilon \downarrow 0, \quad (1)$$

where a is the positive Weyl symbol. Additional hypotheses can of course exclude the example; the source paper's own proved theorem does so.

2 Definitions and idea of the proof

We use the source's 2π convention:

$$(\text{Op}^W(a)f)(x) = \int_{\mathbb{R}^{2d}} a\left(\frac{x+y}{2}, \omega\right) e^{2\pi i(x-y)\cdot\omega} f(y) dy d\omega. \quad (2)$$

For a compact operator on a real Hilbert space, $H_T(\varepsilon)$ is the logarithm of the least number of closed ε -balls covering $T(B)$. The spreading function is $\eta = \mathcal{F}_s a$; the symplectic permutation does not affect a radial Gaussian or its L^1 norm.

Idea of the proof. An integrable spreading function controls operator norm, not spectral rank. A Gaussian rank-one projector has only one entropy axis. Its strictly positive Gaussian Weyl symbol, however, exceeds ε on a $2d$ -dimensional ball of radius comparable to $\sqrt{\log(1/\varepsilon)}$. Integrating the logarithmic height over this ball yields order $(\log(1/\varepsilon))^{d+1}$, whereas the one-axis entropy has order $\log(1/\varepsilon)$.

3 Counterexample

Theorem 1. *For every integer $d \geq 1$ and every $\delta > 0$, there is a positive, self-adjoint, rank-one Weyl operator T on real $L^2(\mathbb{R}^d)$ whose Weyl symbol a is strictly positive and Schwartz, and whose spreading function η is Schwartz with $\|\eta\|_1 < \delta$, such that*

$$\frac{H_T(\varepsilon)}{\int_{\mathbb{R}^{2d}} \log_+(a(z)/\varepsilon) dz} \longrightarrow 0.$$

In particular, (1) fails even at arbitrarily small spreading L^1 norm.

Proof. Let

$$g(t) = 2^{d/4} e^{-\pi|t|^2}, \quad T_c f = c \langle f, g \rangle g,$$

where $c > 0$. Since $2^{d/2} \int_{\mathbb{R}^d} e^{-2\pi|t|^2} dt = 1$, the vector g is normalized. Thus T_c is positive, self-adjoint, and rank one.

Its integral kernel is $K_c(x, y) = cg(x)g(y)$. Put $u = (x + y)/2$ and $v = x - y$. The inverse of (2) at the kernel level gives

$$\begin{aligned} a_c(u, \omega) &= \int_{\mathbb{R}^d} K_c(u + v/2, u - v/2) e^{-2\pi i v \cdot \omega} dv \\ &= c 2^{d/2} e^{-2\pi|u|^2} \int_{\mathbb{R}^d} e^{-\pi|v|^2/2} e^{-2\pi i v \cdot \omega} dv \\ &= c 2^d e^{-2\pi(|u|^2 + |\omega|^2)}. \end{aligned}$$

This is strictly positive and Schwartz. The $2d$ -dimensional Gaussian Fourier formula, with the harmless symplectic permutation, yields

$$\eta_c(\zeta) = c e^{-\pi|\zeta|^2/2}, \quad \|\eta_c\|_1 = c 2^d. \quad (3)$$

Choosing $0 < c < \delta/2^d$ proves the asserted arbitrary smallness.

On the real unit ball, $\langle f, g \rangle$ ranges over $[-1, 1]$, and hence $T_c(B) = [-c, c]g$. Projection of any covering center onto the line $\mathbb{R}g$ cannot increase its distance to this interval. Covering an interval of length $2c$ by intervals of radius ε therefore gives, for $0 < \varepsilon < c$,

$$N(\varepsilon; T_c(B)) = \left\lceil \frac{c}{\varepsilon} \right\rceil, \quad H_{T_c}(\varepsilon) = \log \left\lceil \frac{c}{\varepsilon} \right\rceil = \log(c/\varepsilon) + o(1). \quad (4)$$

It remains to compute the proposed right side. Set $L = \log(c2^d/\varepsilon)$ and suppose $0 < \varepsilon < c2^d$. The positive part is supported on the ball $|z|^2 < L/(2\pi)$ in $\mathbb{R}^{2d}$. If $v_{2d} = \pi^d/d!$ is its unit-ball volume and $R^2 = L/(2\pi)$, radial integration gives

$$\begin{aligned} I_c(\varepsilon) &= \int_{|z| < R} (L - 2\pi|z|^2) dz \\ &= Lv_{2d}R^{2d} - 2\pi \frac{2d}{2d+2} v_{2d}R^{2d+2} \\ &= \frac{L^{d+1}}{2^d d! (d+1)}. \end{aligned} \quad (5)$$

Equations (4) and (5) imply $H_{T_c}(\varepsilon)/I_c(\varepsilon) \rightarrow 0$ for every $d \geq 1$. □

4 A separate obstruction to the commented inversion route

Proposition 1. *Suppose an operator has a spreading representation*

$$S = \int_{\mathbb{R}^{2d}} \eta(z) \pi(z) dz$$

in the weak sense, where the time-frequency shifts $\pi(z)$ are unitary and $\eta \in L^1$. Then S is bounded and $\|S\| \leq \|\eta\|_1$ (up to the fixed normalization constant of the chosen spreading convention). In particular, its eigenvalues cannot tend to infinity.

Proof. For $f, h \in L^2$, unitarity gives

$$|\langle Sf, h \rangle| \leq \int |\eta(z)| |\langle \pi(z)f, h \rangle| dz \leq \|\eta\|_1 \|f\|_2 \|h\|_2.$$

Taking the supremum over unit vectors proves the norm bound. □

The official source comments place finite spreading L^1 norm next to an operator “whose eigenvalues tend to infinity” before proposing inversion. Proposition 1 shows that these conditions cannot describe one fixed operator under the standard spreading representation. A coherent replacement would require a scaling family, another unbounded class, or an independent spectral-counting hypothesis.

5 Verification, limitations, and novelty

The included standard-library script numerically evaluates the radial integral in 45 combinations of d, c , and ε , checks the arbitrary-norm scaling in (3), and checks decay of the entropy/integral ratio. It returned **PASS**. These tests guard arithmetic and normalization; the proof above is exact and does not depend on computation.

The Gaussian spreading function is not compactly supported. Therefore the packet makes no claim against definitions of underspreadness that require compact spreading support of area below a threshold. Nor does it contradict Theorem 1 of the source: a_c is not a negative-order hypoelliptic symbol with the stated regularly varying volume. The result instead proves that integrable spreading alone cannot replace the source theorem’s spectral regularity mechanism.

A bounded novelty search on 11 August 2026 used the exact future-work phrase, the target title and authors, and combinations of *metric entropy*, *underspread*, *log integral*, *rank one*, and *Wigner*. It found general approximate-diagonalization and scaled Szegő-asymptotic literature, but no exact Gaussian rank-one entropy obstruction or later answer to this 2026 direction. This is not an exhaustive novelty certification.

Human-review recommendation. Check first that the source-comment finite- L^1 formulation is the intended scope. Then verify the Weyl normalization in the kernel calculation and the compact-support exclusion. The entropy and radial integration steps are elementary.

References

- [1] T. Allard and H. Bölcskei, *Entropy and Minimax Risk of Hypoelliptic Pseudodifferential Operators*, arXiv:2603.23744, 2026.